Topology, size stabilization and action normalization

A fixed nonzero topological sector can have a positive classical energy floor while admitting arbitrarily small size-based action. Quadratic/quartic gradient competition can stabilize size, but then its coefficient b/c already has action units; the common stiffness normalization remains free. Neither model has a vacuum frequency gap or derives a quantum phase rule. The two explicit families below test different hypotheses: actual two-dimensional static solutions, and a three-dimensional trial family with a preferred radius.

The sector bounds, sigma/Skyrme ingredients and Derrick scaling have established prior art. These are written conditional constructions with the source reading levels stated below; historical checks were internal. No full field-minimizer or global evolution theorem is inferred from a one-parameter scale minimum.

A sector energy floor with actual solutions at every radius

Take $n(t,x) \in S^2$, $x \in \mathbb{R}^2$, with stiffness $\rho > 0$ in energy units, speed $c > 0$ and physical Lorentzian action

$$S[n] = \frac{\rho}{2}\int dt\, d^2x[c^{-2}|\partial_t n|^2 - |\nabla n|^2], \qquad H = \frac{\rho}{2}\int [c^{-2}|\partial_t n|^2 + |\nabla n|^2]d^2x.$$

The canonical momentum density is $\pi = \rho\partial_t n/c^2$. Require finite energy and a smooth one-point compactification with $n(\infty) = e_3$. Degree

$$Q = \frac{1}{4\pi}\int n\cdot(\partial_1 n \times \partial_2 n)d^2x \in \mathbb{Z}$$

is conserved during smooth evolutions preserving this boundary condition. Restrict to $|Q| = 1$; the model does not explain exclusion of $Q = 0$. For $a = \partial_1 n$, $b = \partial_2 n$, $q = n\cdot(a\times b)$,

$$|a \pm n\times b|^2 = |a|^2 + |b|^2 \mp 2q$$

proves $H \geq E \geq 4\pi\rho|Q|$. For each $R > 0$ the explicit field

$$n_R(x) = \frac{(2Rx_1, 2Rx_2, r^2 - R^2)}{r^2 + R^2}, \qquad r = |x|,$$

has unit norm and boundary e_3, and direct differentiation gives

$$|\nabla n_R|^2 = \frac{8R^2}{(r^2+R^2)^2}, \qquad q_R = -\frac{4R^2}{(r^2+R^2)^2}.$$

Using $d^2x = 2\pi r\, dr$ and $\int_0^\infty r(r^2+R^2)^{-2}dr = 1/(2R^2)$ gives $Q = -1$ and $E = 4\pi\rho$. Reflection reverses degree. Saturating the sector minimum makes each a stationary solution, not just a variational trial. Its enclosed energy is

$$E_R(r) = 4\pi\rho\frac{r^2}{r^2+R^2},$$

so R is its half-energy radius. As $R \downarrow 0$, every member stays smooth in the sector, while the limit is singular at the origin and its energy measure tends weakly to $4\pi\rho\delta_0$. Gradients do not converge strongly to the vacuum in L^2. Excluding the singular endpoint does not exclude this family of preparations; no dynamical collapse or finite-inertia dilation mode is claimed.

Which action and which time are tested?

For the specified crossing duration $\tau_R = R/c$, a static solution has

$$\mathcal{A}_R = |S[n_R; 0, R/c]| = E[n_R]R/c = 4\pi\rho R/c \longrightarrow 0.$$

This is an energy-times-crossing-time observable, not a period of oscillation or every possible action functional. At fixed external duration T the same solutions instead have $|S| \geq 4\pi\rho T$, including that supplied clock. Their canonical motion cost $\int \pi \cdot \partial_t n$ is identically zero. Under spacetime dilation $n_\lambda(t, x) = n(t/\lambda, x/\lambda)$ with scaled duration, $S[n_\lambda] = \lambda S[n]$, degree and the field equation being preserved at fixed ρ, c. Those constants alone supply no action unit. A radius floor $R \geq \ell$ would add $\mathcal{A}_R \geq 4\pi\rho\ell/c$ within this family, using a new length.

Near the vacuum, $n = (u_1, u_2, \sqrt{1 - |u|^2})$ gives $\partial_t^2 u = c^2\Delta u$, $\omega(k) = c|k|$. On $L^2(\mathbb{R}^2; \mathbb{R}^2)$, the squared-frequency operator $-c^2\Delta$ with domain H^2 has spectrum $[0, \infty)$. Normalized packets $u_L = L^{-1}f(x/L)$ have Rayleigh quotient $c^2L^{-2}\|\nabla f\|_2^2/\|f\|_2^2 \to 0$. A periodic box with constant rotations removed has first frequency $2\pi c/L$, which closes with volume. Within the degree-one sector, translations $n_R(x - a)$ give square-integrable zero tangents of the static second variation. No positive internal gap after quotienting them is asserted. Sector energy, classical frequency and quantum Hamiltonian spectrum differ.

A stabilized three-dimensional trial family

Now use $n : \mathbb{R}^3 \to S^3 \subset \mathbb{R}^4$, approaching a fixed point at infinity, with degree N, and static energy

$$E[n] = \int \left[a \sum_i |\partial_i n|^2 + b \sum_{i<j} (|\partial_i n|^2 |\partial_j n|^2 - (\partial_i n \cdot \partial_j n)^2) \right] d^3x.$$

Here $a > 0$ has units energy/length, $b > 0$ energy times length. Specify physical time by the conservative completion

$$L = \int \left[\frac{a}{c^2} |\dot{n}|^2 + \frac{b}{c^2} \sum_i (|\dot{n}|^2 |\partial_i n|^2 - (\dot{n} \cdot \partial_i n)^2) \right] d^3x - E[n].$$

Its velocity quadratic form is positive; this is the quadratic/quartic Lorentz completion with $x^0 = ct$. Smooth solutions are assumed where they exist; no global nonlinear well-posedness or hard-speed theorem for arbitrary backgrounds is supplied.

If λ_i are the singular values of dn, pairing $a\lambda_i^2$ with $b\lambda_j^2\lambda_k^2$ gives static density at least $6\sqrt{ab}\lambda_1\lambda_2\lambda_3$. The absolute Jacobian integral bounds absolute degree and $\operatorname{vol} S^3 = 2\pi^2$, so

$$E \geq 12\pi^2\sqrt{ab}|N|.$$

For $y = x/R$, $r = |y|$, the inverse stereographic trial profile is

$$n_R(x) = \frac{(2y_1, 2y_2, 2y_3, r^2 - 1)}{1 + r^2}, \quad |N| = 1.$$

Its three dimensionless singular values are $2/(1 + r^2)$, giving

$$A = 48\pi \int_0^\infty \frac{r^2 dr}{(1 + r^2)^2} = 12\pi^2, \qquad B = 192\pi \int_0^\infty \frac{r^2 dr}{(1 + r^2)^4} = 6\pi^2.$$

The substitution $r = \tan u$ proves both integrals. Scaling then gives

$$E(R) = aAR + bB/R, \quad R_* = \sqrt{b/(2a)}, \quad E(R_*) = 12\sqrt{2}\pi^2\sqrt{ab}.$$

The scale minimum is unique with second derivative $2bB/R^3 > 0$; both shrinking and expansion cost energy. Shape variations are outside this calculation. In particular n_{R_*} is not asserted to solve the static Euler–Lagrange equation.

The remaining common normalization

For an off-shell static trial field over $T_R = R/c$, the diagnostic action is

$$\mathcal{A}_R = E(R)R/c = (aAR^2 + bB)/c.$$

Its infimum within this family is $6\pi^2 b/c$, and at the energy-optimal radius it is $12\pi^2 b/c$. The coefficient b/c already has action units. Rescaling $x = \ell y$, $t = \ell s/c$, $\ell = \sqrt{b/a}$, gives the **full** action, including kinetic terms,

$$S = (b/c)\widetilde{S}[n].$$

Replacing (a, b) by $(\eta a, \eta b)$, $\eta > 0$, leaves the length, classical field equations, degree and vacuum speed unchanged, while energy and action scale by η. This varies physical stiffnesses; it does not deny a bound with both coefficients fixed. No degenerate $\eta = 0$ model is needed for the zero-infimum countertest. At quadratic order around a vacuum,

$$L^{(2)} = a \int [c^{-2}|\dot{\pi}|^2 - |\nabla\pi|^2] d^3x, \qquad \Omega(k) = c|k|.$$

The quartic term has no quadratic vacuum contribution. Stabilizing a nonzero-degree radius therefore leaves vacuum waves gapless and does not select that sector or an absolute action normalization.

Source evidence and consequence for STATE

Ody–Ryder 1994, §2, pp. 4–7, equations (2.1)–(2.26), was read for the static degree, square completion and arbitrary-scale ingredients. This is a static source match; it does not supply the Lorentzian kinetic term or crossing-time prescription. The Lorentzian time law, half-energy crossing diagnostic and spectrum comparison are the specified model tests here. The generalized Skyrme-bound 2014 source was read at abstract level only; no full-text formula match is claimed. The coefficients, trial integrals and normalization above have written derivations.

Topology supplies a conditional sector-energy floor; gradient competition can supply size at fixed coefficients. Neither determines physical $\hbar$, quantum states/instruments or a universal positive action. The separate reciprocal-coupling test asks what interaction fixes about relative normalizations.